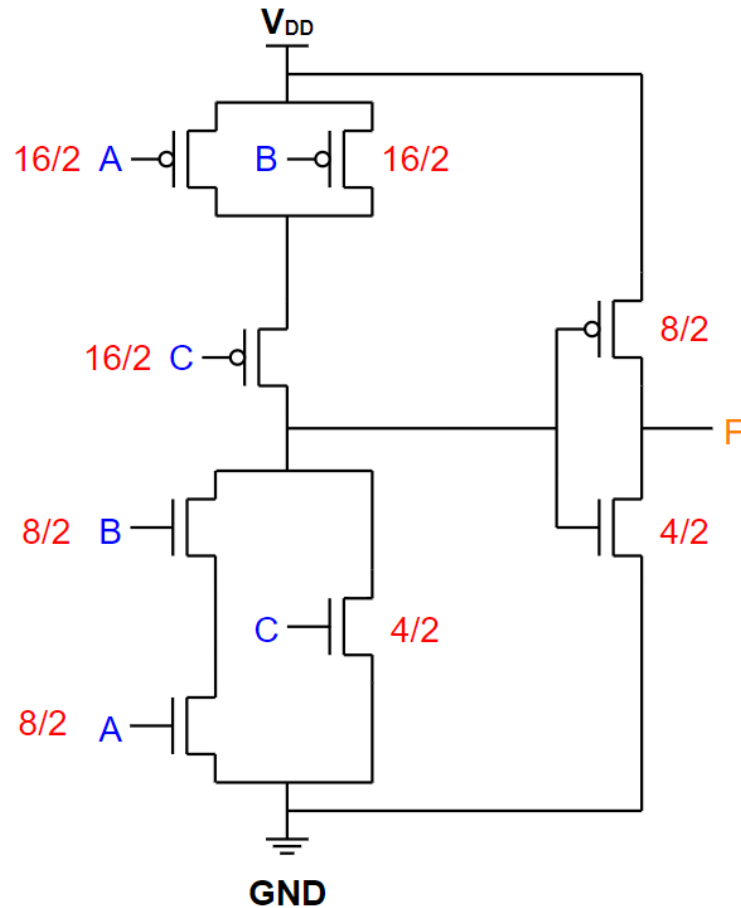
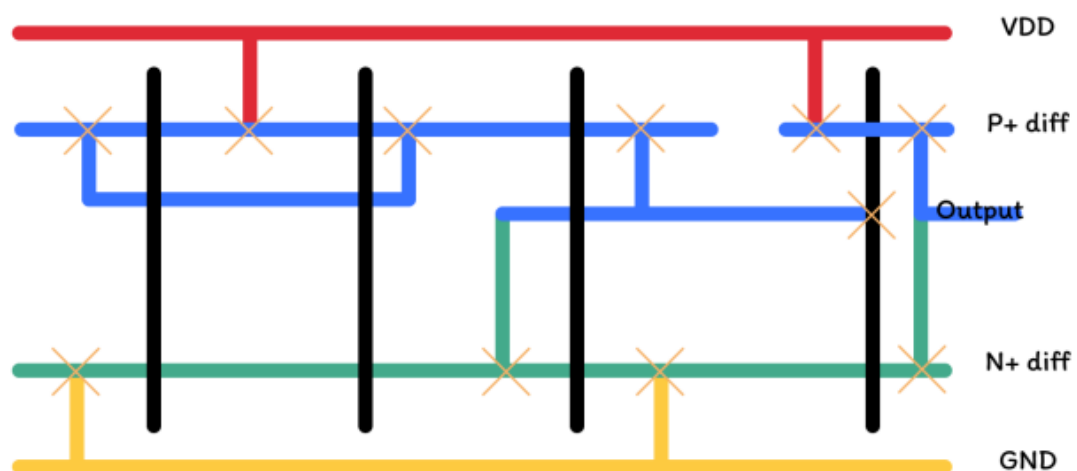


➤ Part (1): $F = AB + C$:

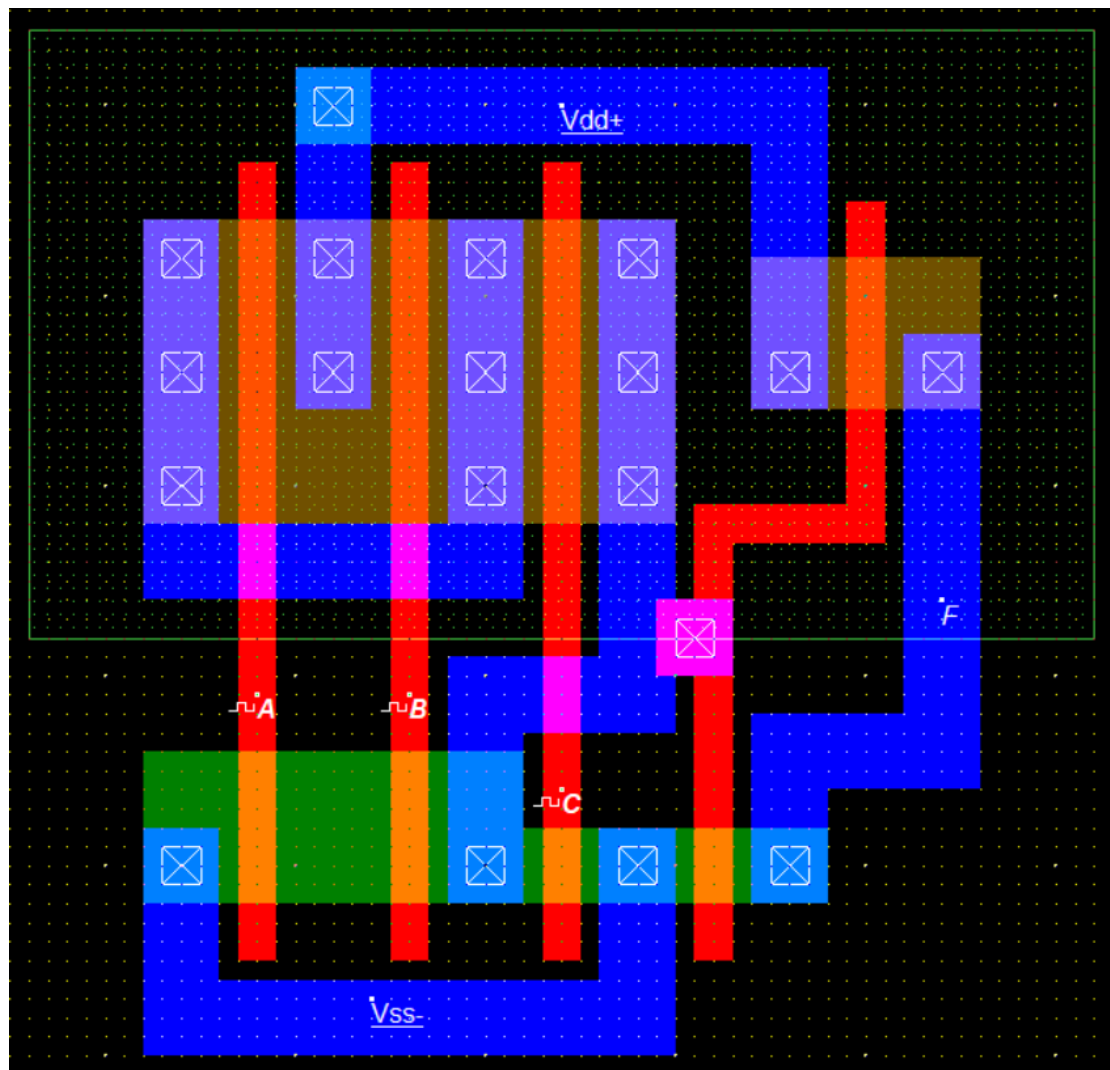
- a. Draw the schematic and size the transistors such that the smallest transistor is $4/2$. Make each stage is equivalent to the Unit Inverter.



- Stick diagram:



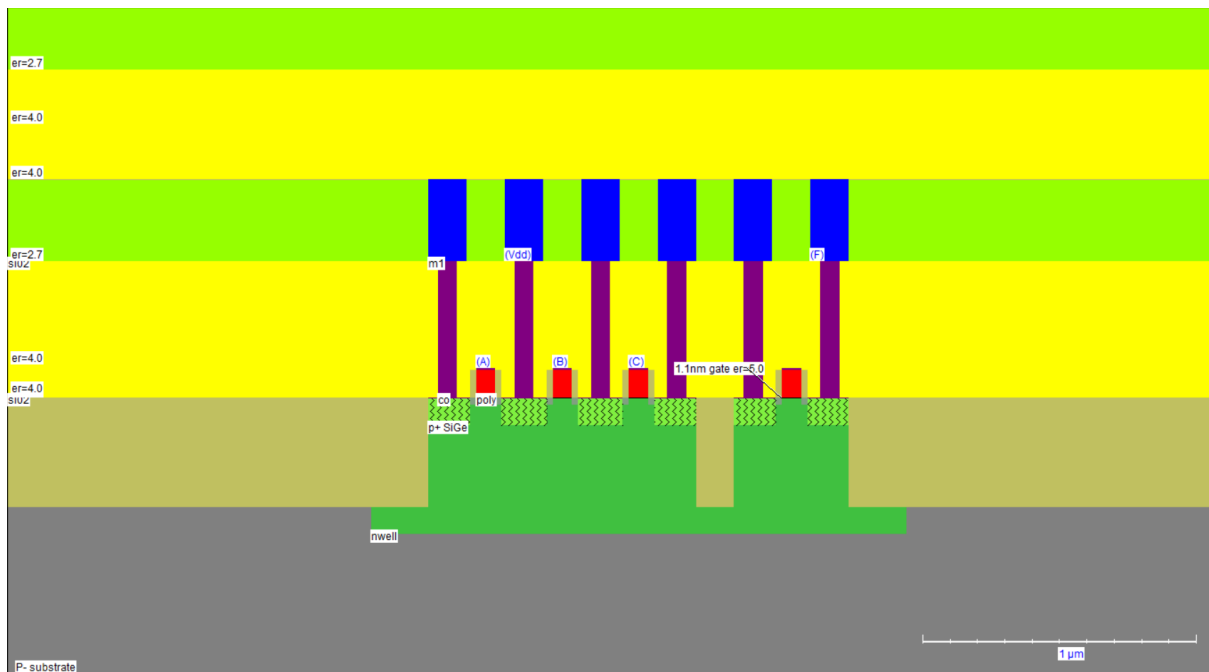
- b. Draw the Layout using Microwind. Make sure you put the Body Contacts.



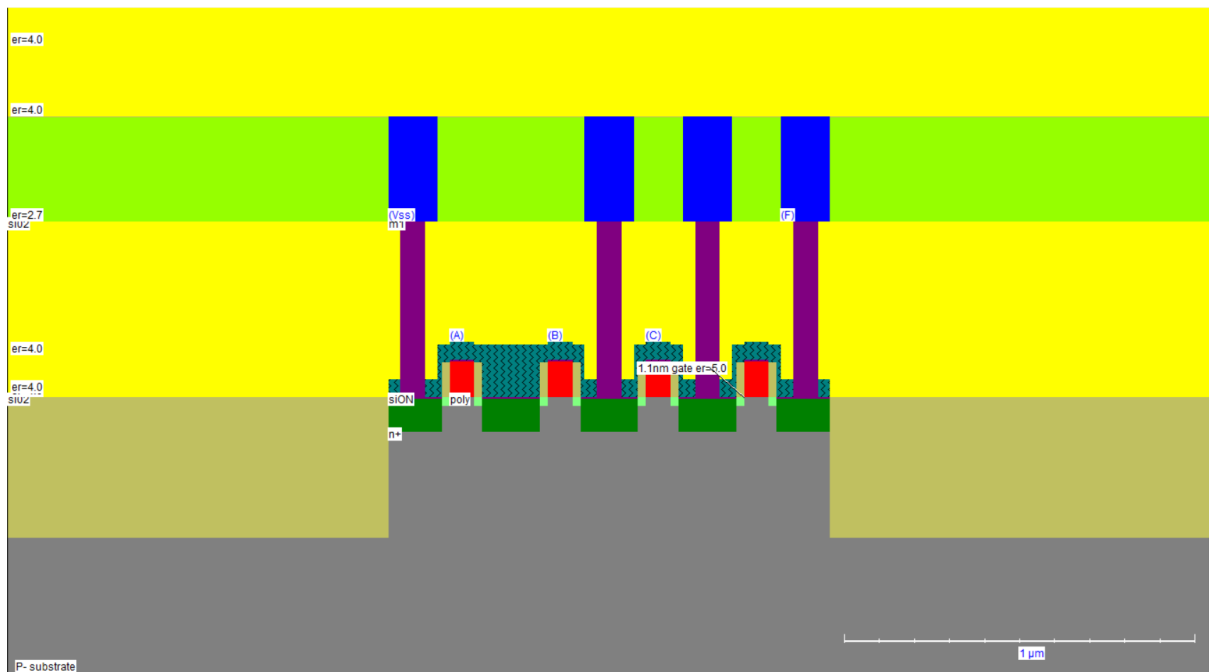
- Truth table:

A	B	C	F
0	0	0	0
1	0	0	0
0	1	0	0
1	1	0	1
0	0	1	1
1	0	1	1
0	1	1	1
1	1	1	1

c. Draw a cross section in the P strip(s) (Use the Snipping tool to extract the image).

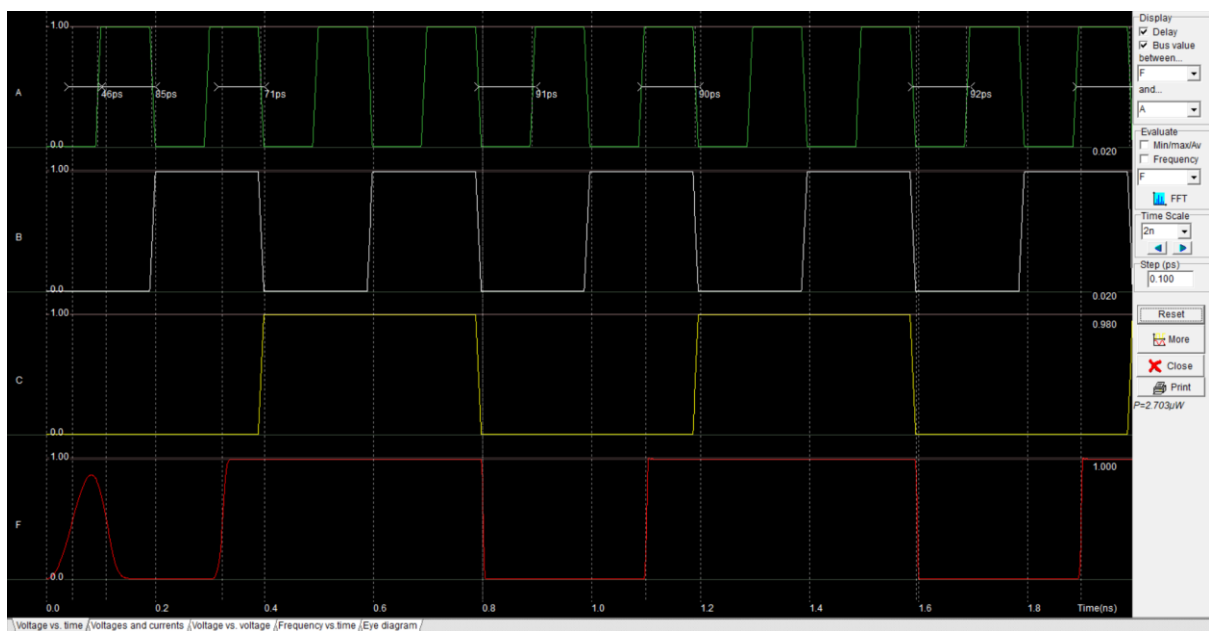


d. Draw a cross section in the N strip(s) (Use the Snipping tool to extract the image).

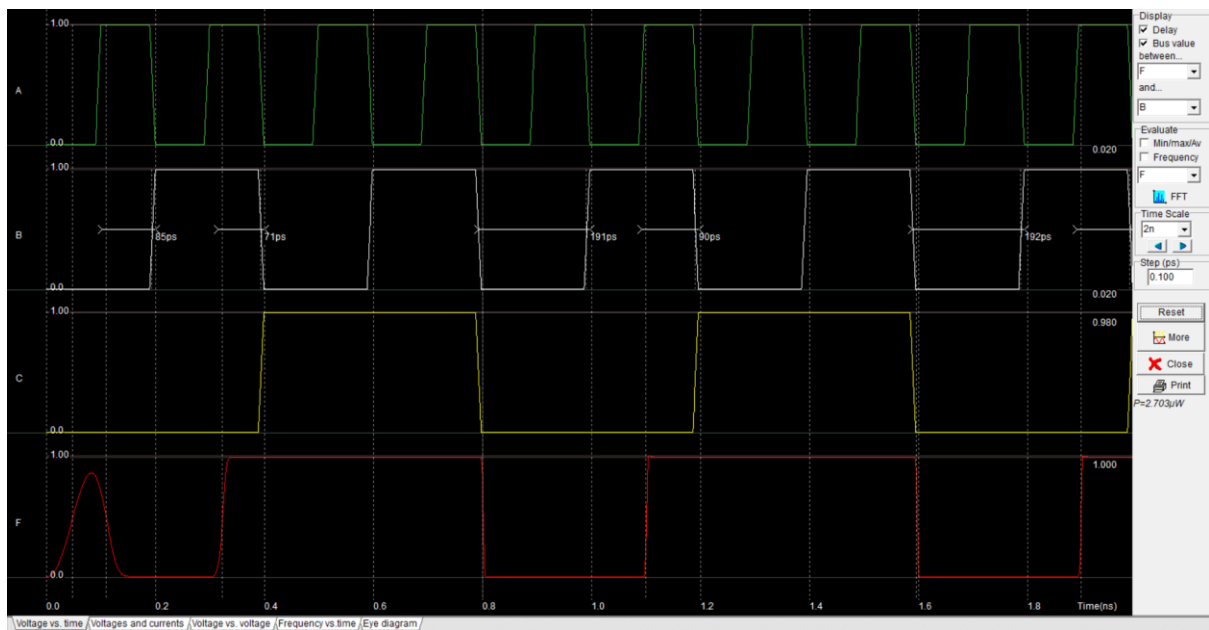
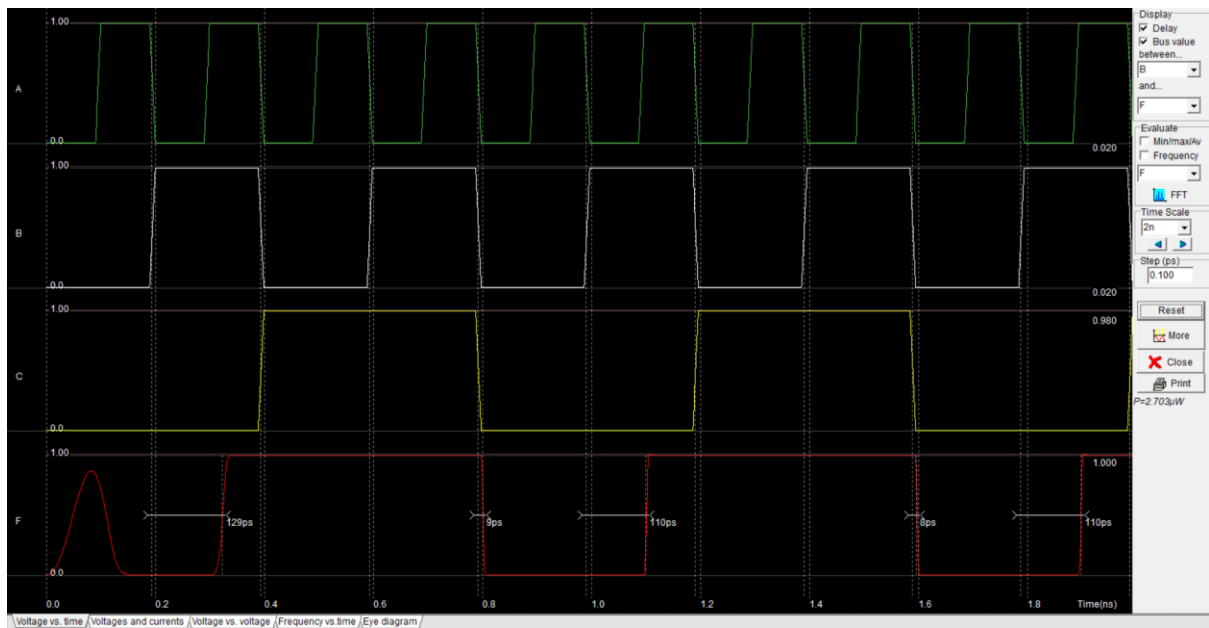


- e. Put clocks at the Inputs A, B and C. Simulate the Circuit and find the delays between the input and the output using the built-in SPICE Simulator. Show the delays (extract an Image using the Snipping Tool).

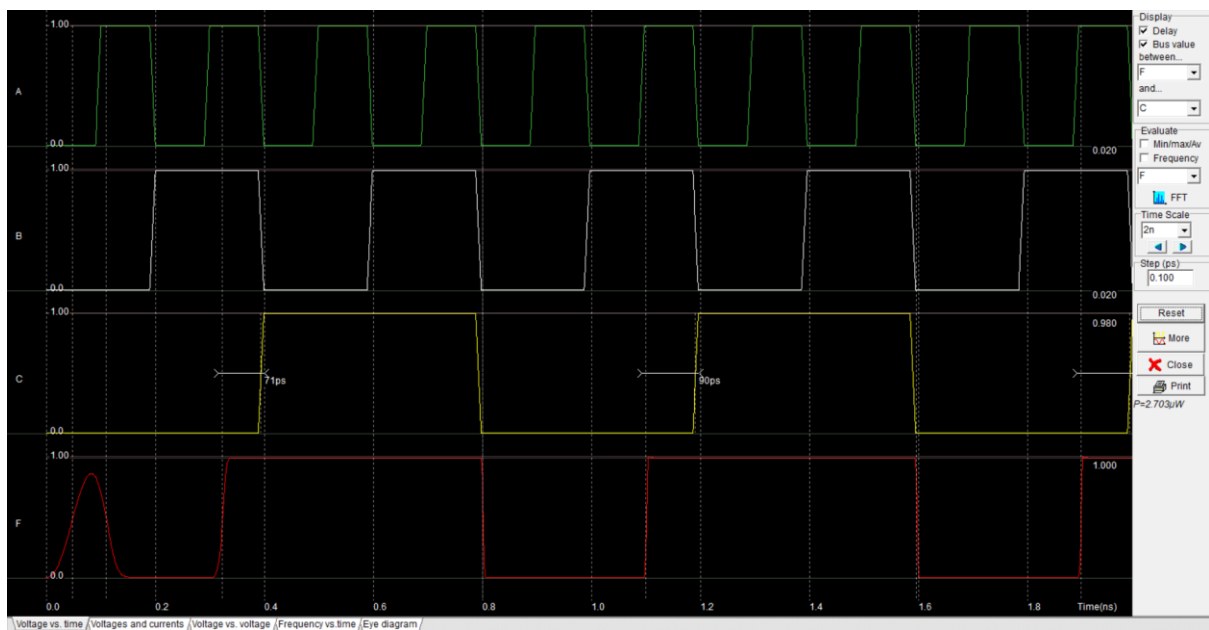
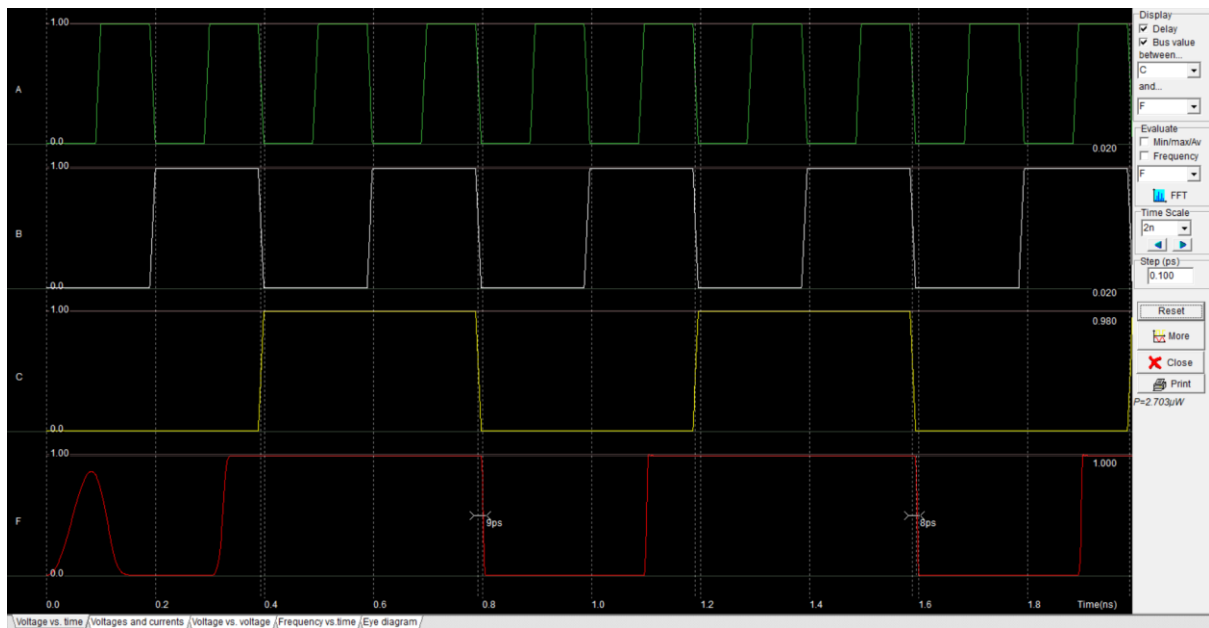
1. Delay between A and F:



2. Delay between B and F:

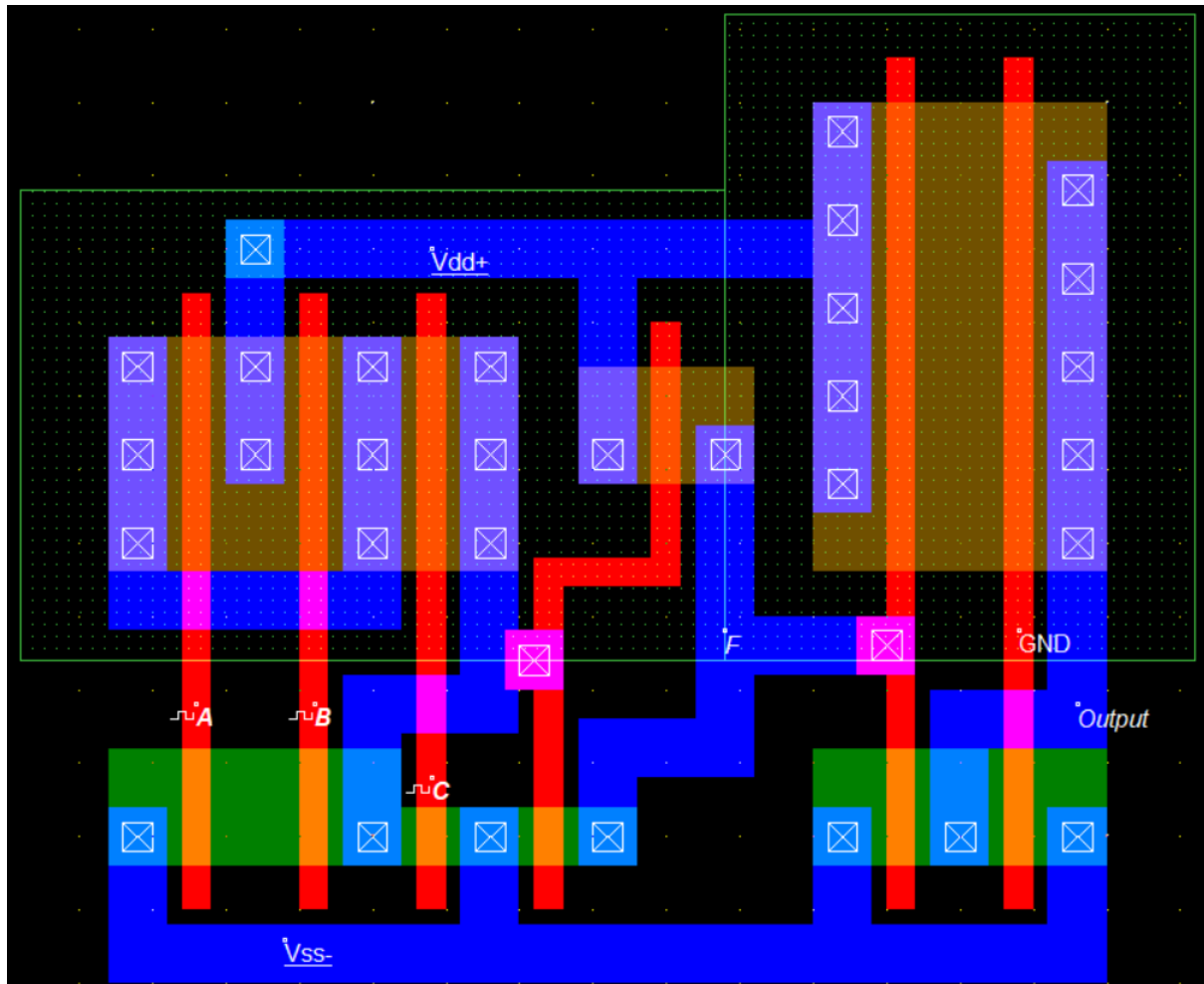


3. Delay between C and F:

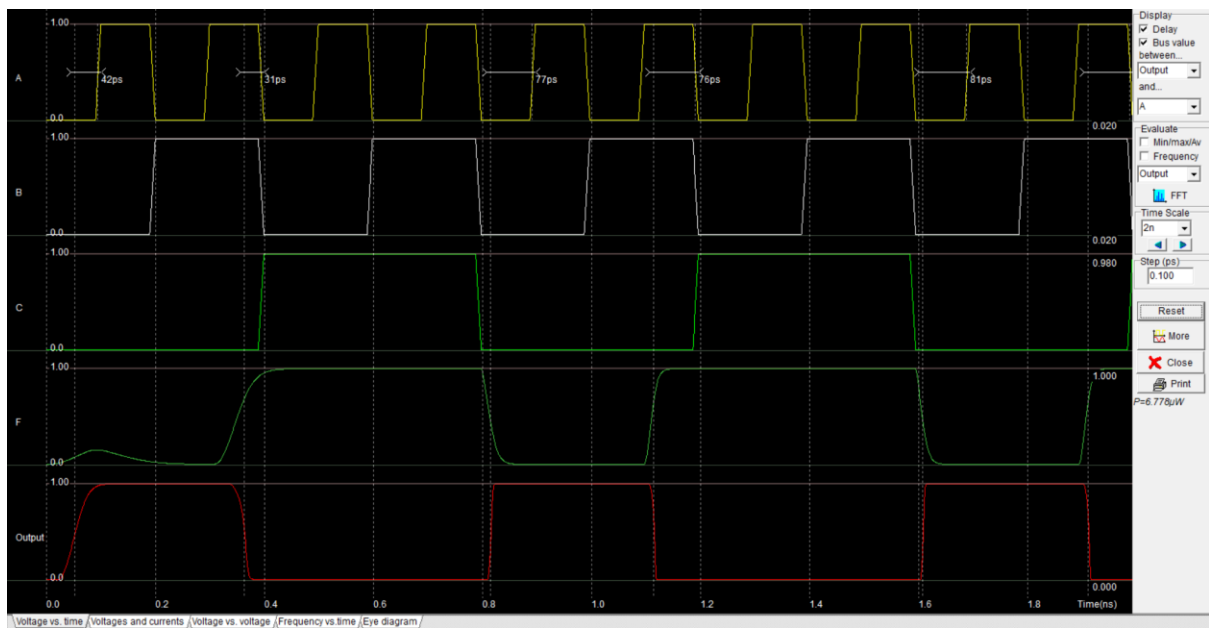
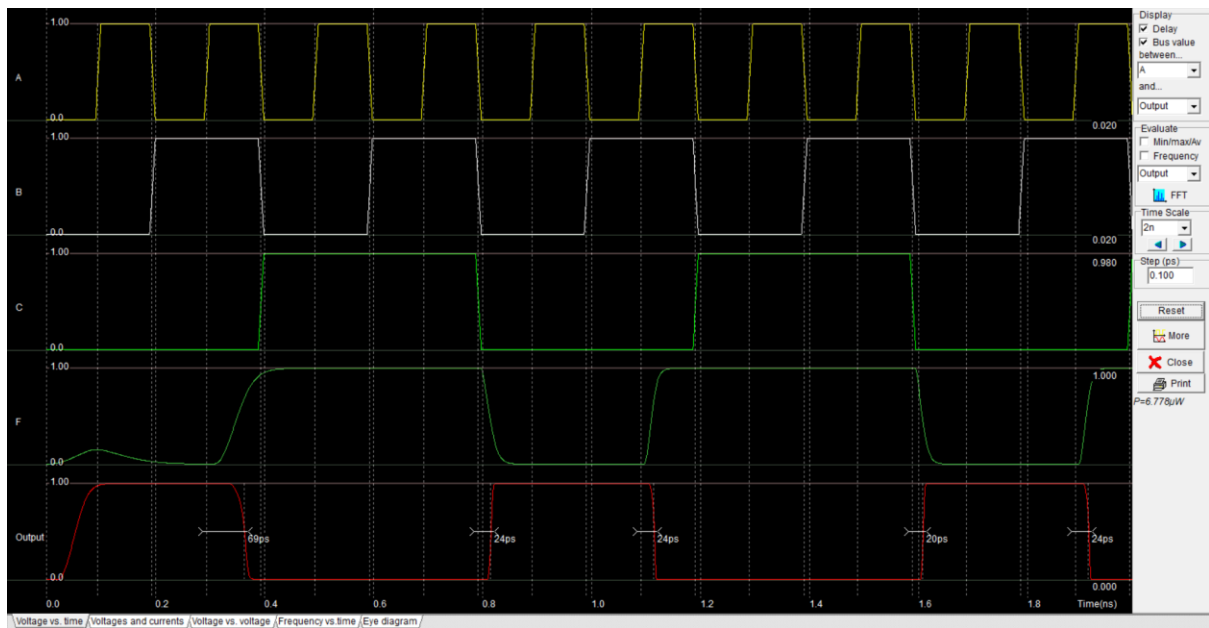


➤ Part (2): Adding a 2-input NOR gate to part (1):

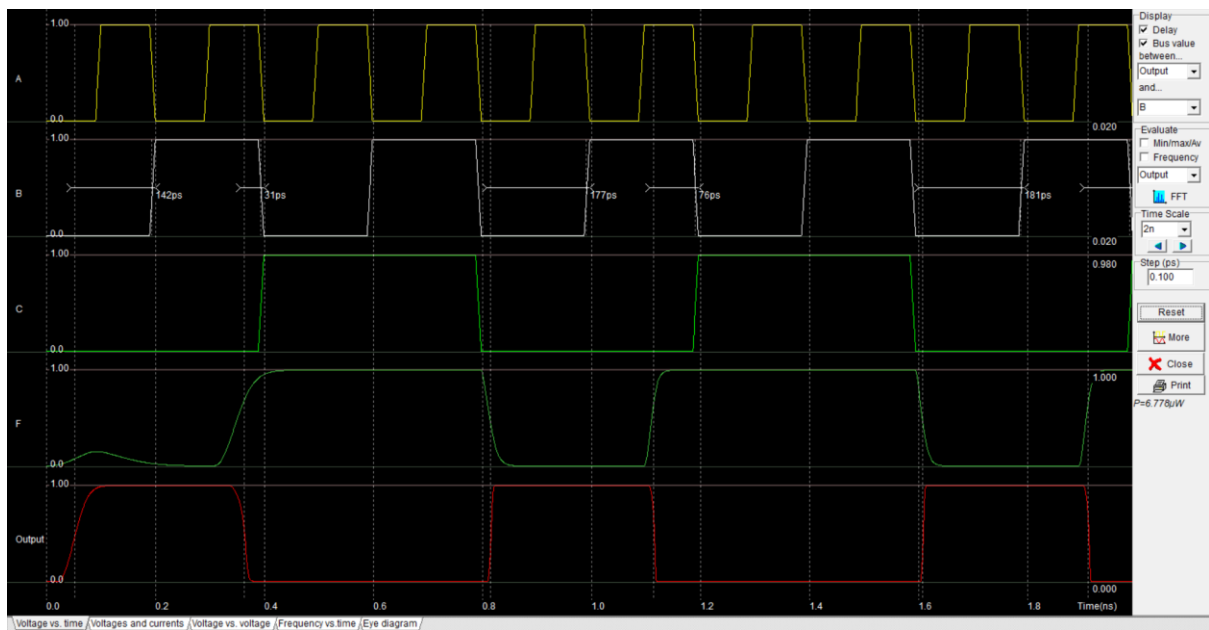
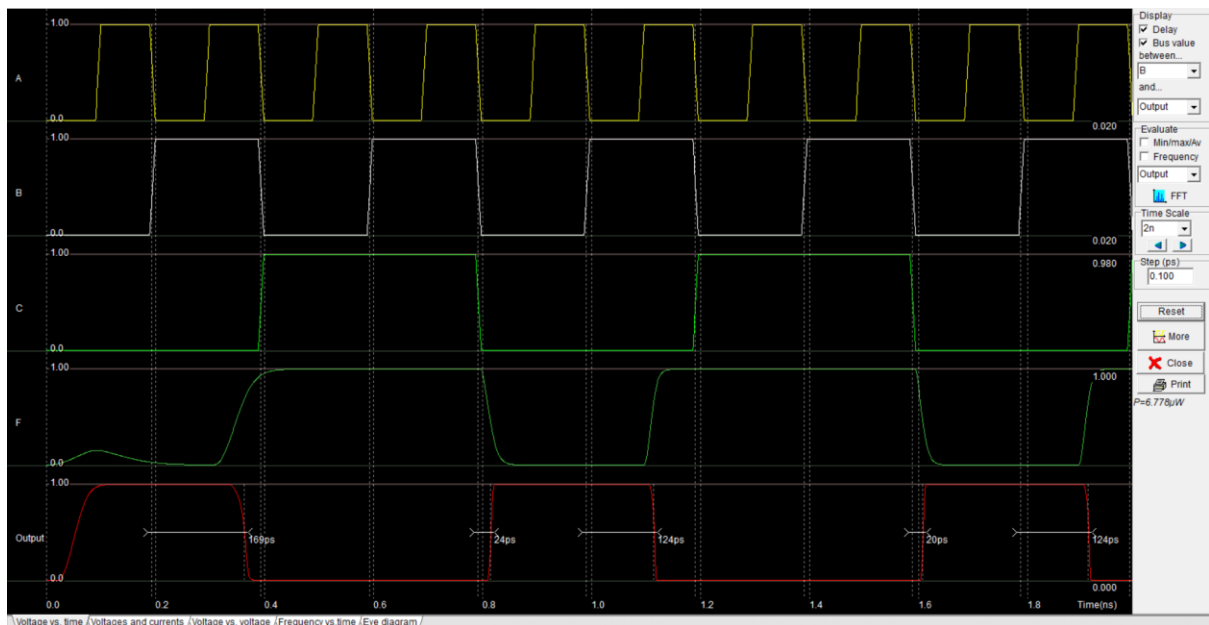
- Make a copy of the file (msk file) now add a 2-input NOR gate to the new file. Make F drive one input the NOR gate. Simulate all the gate and show the delay of whole circuit. Label the other input GND on it Make the Size of each N transistor 2 units = (8/2).



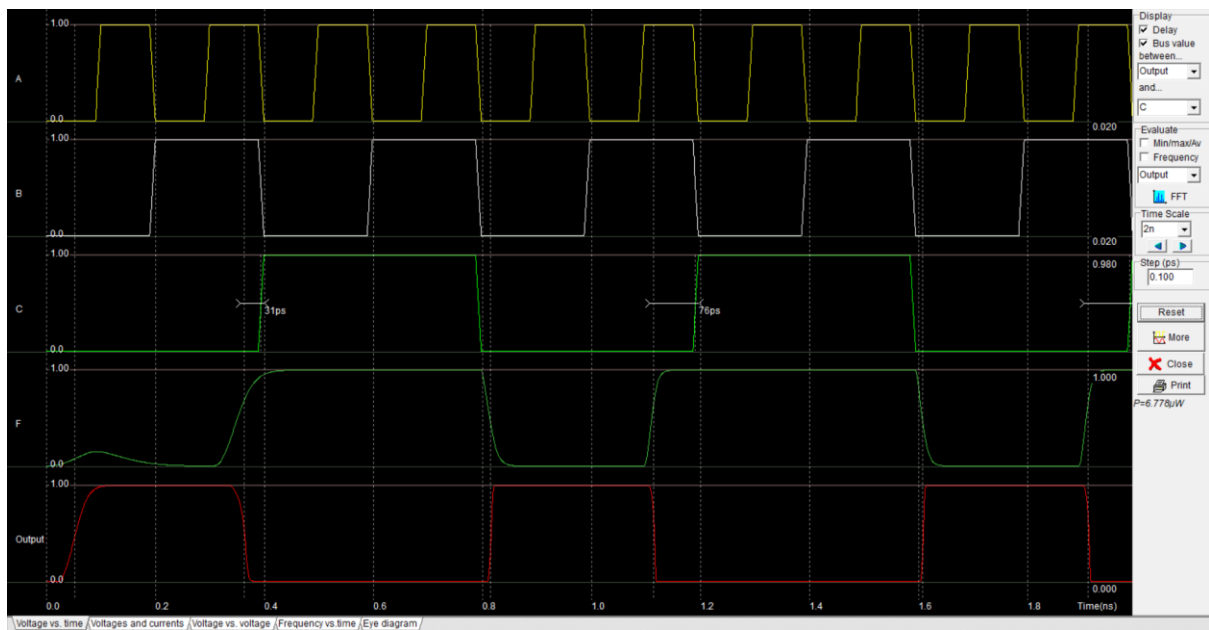
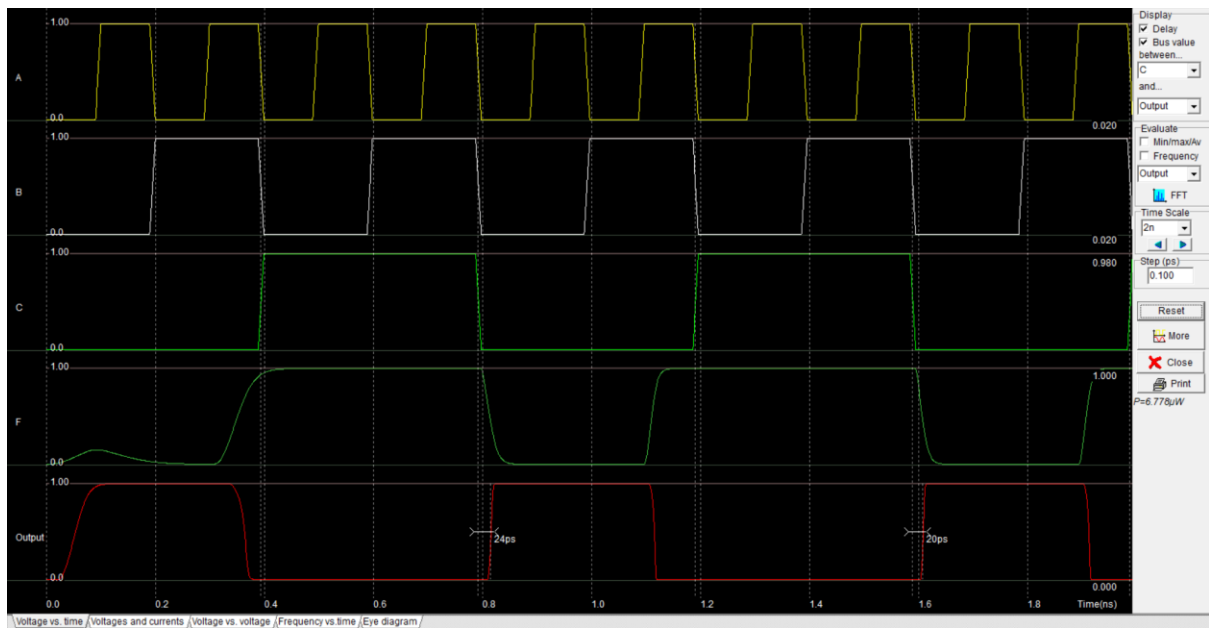
1. Delay between A and Output:



2. Delay between B and Output:



3. Delay between C and Output:



4. Delay between F and Output:

